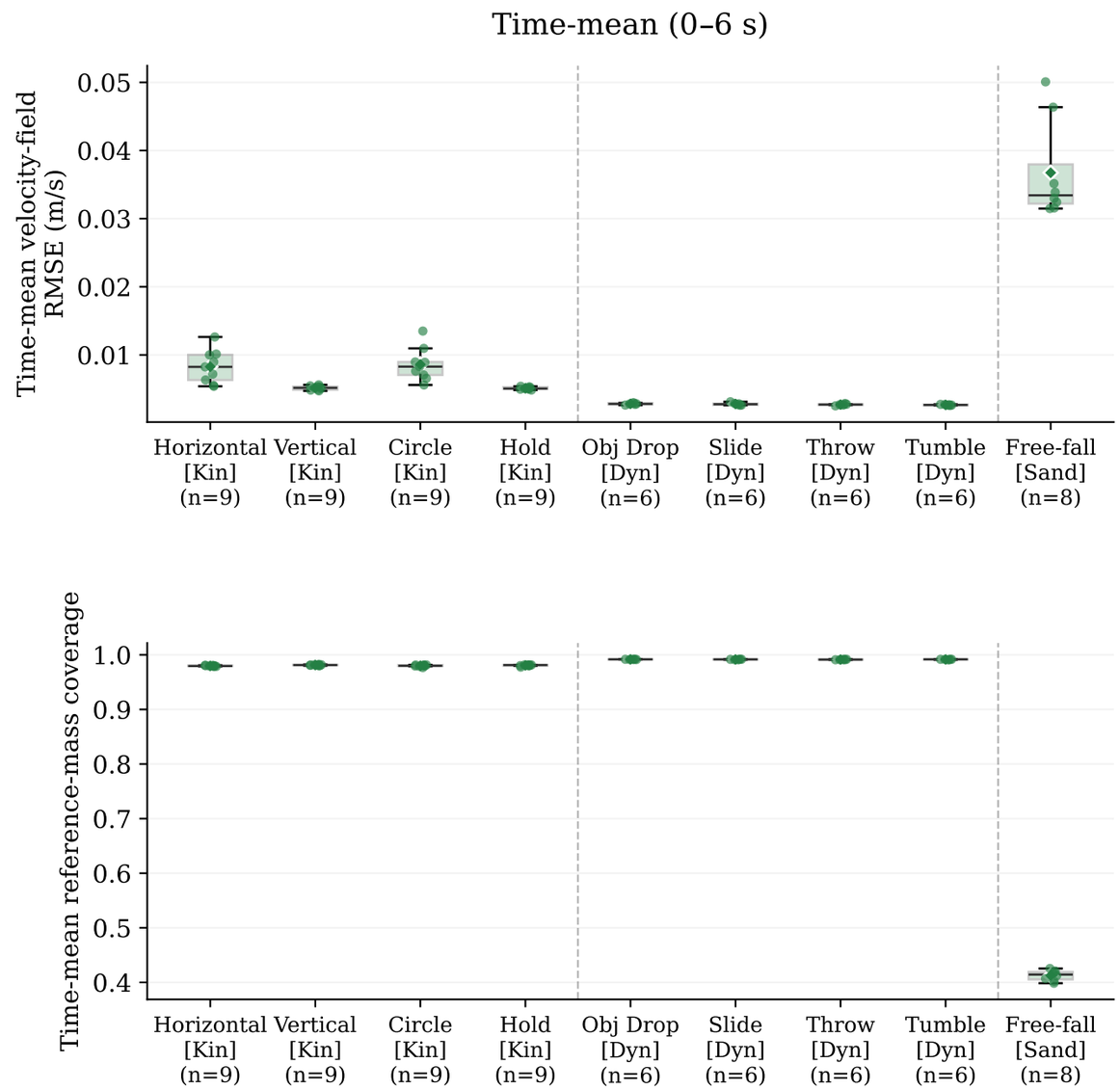
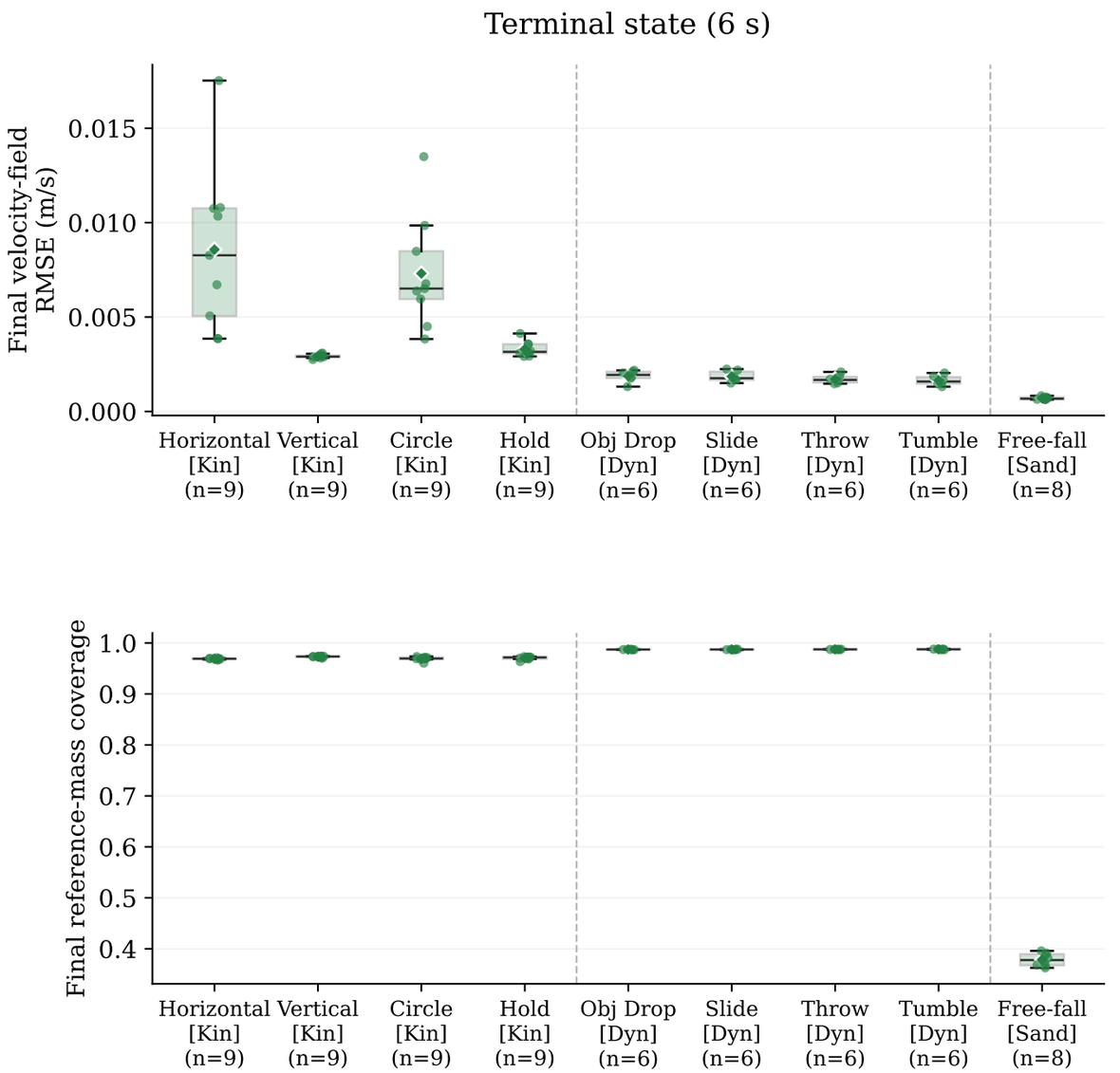
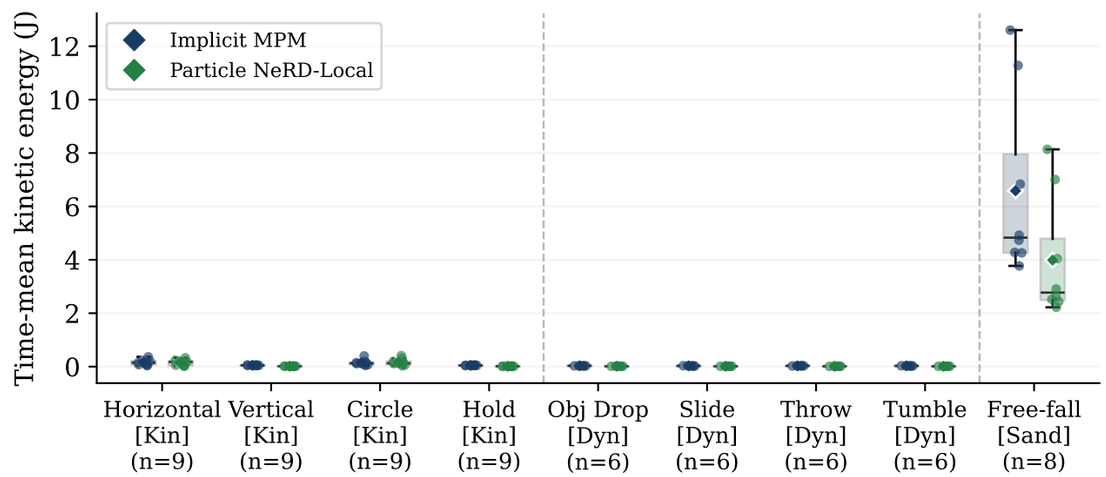
By Motion Type — Kinematic Object Motions (Horizontal, Vertical, Circle, Hold) | Dynamic Object Motions (Obj Drop, Slide, Throw, Tumble) | Sand Free-Fall (Flat Ground)







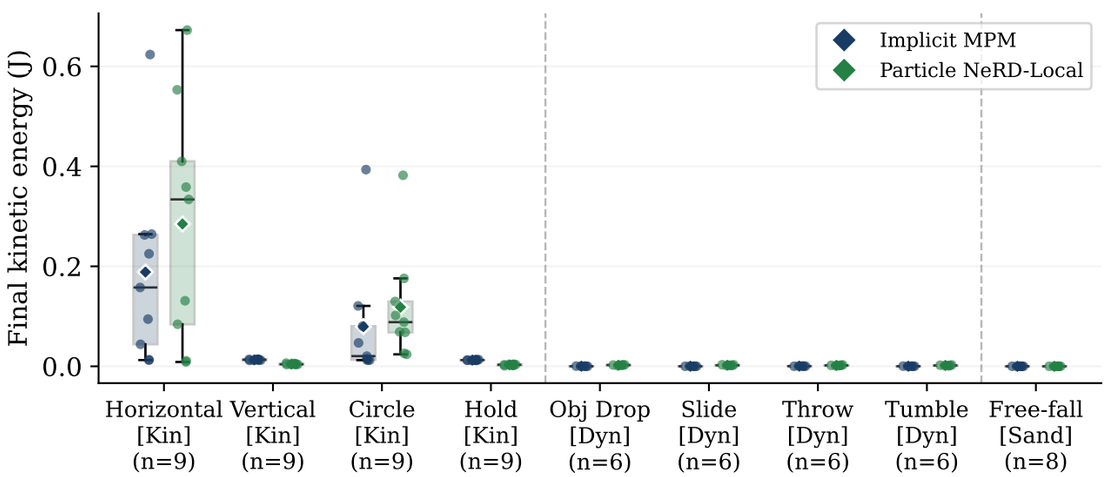


Figure 4b: Frozen object-model checkpoint 10,800. Velocity-field RMSE, reference-mass coverage, and particle kinetic energy grouped by motion: Kinematic object motions (Horizontal, Vertical, Circle, Hold), Dynamic object motions (Obj Drop, Slide, Throw, Tumble), and Unconfined Sand Free-Fall on flat ground.